
CHAPTER 1

TRICKERY

1 Some Telescoping Trickery

1.1 Some easy telescopes

Example : $1(1!) + 2(2!) + \cdots + n(n!) = (n+1)! - 1$

Simple application of **telescoping** trick. Write $j(j!) = (j+1-1)j! = (j+1)j! - j! = (j+1)! - j!$ which gives us $s = \sum_{j=1}^n (j)(j!) = \sum_{j=1}^n (j+1)! - j!$ which is just $(2! - 1!) + (3! - 2!) + \cdots + (n! - (n-1)!) + (n+1! - n!) = (n+1)! - 1$ ■

Example : $1/2! + 2/3! + 3/4! + \cdots n - 1/n!$

We rewrite it as

$$(2-1)/2! + (3-1)/3! + (4-1)/4! + \cdots (n-1)/n!$$

whence we get

$$(2)/2! + (3)/3! + \cdots (n)/n! - (1/2! + 1/3! + \cdots 1/n!)$$

which gives us

$$1/1! + 1/2! + \cdots 1/(n-1)! - (1/2! + 1/3! + \cdots 1/(n-1)! + 1/n!) = \\ 1 - 1/n!$$
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1.2 More telescopes

Example : $(1 - 1/4)(1 - 1/9)(1 - 1/16) \cdots (1 - 1/n^2) = \prod_{i=1}^n (1 - 1/i^2)$

Look at $(1 - (1/i)^2) = (1 - 1/i)(1 + 1/i) = ((i-1)/i)((i+1)/i)$ Which means we have $\prod_{i=1}^n \left(\frac{i-1}{i}\right)\left(\frac{i+1}{i}\right) = \frac{2-1}{2} \frac{2+1}{2} \frac{3-1}{3} \frac{3+1}{3} \frac{4-1}{4} \frac{4+1}{4} \cdots \frac{n-1-1}{n-1} \frac{n-1+1}{n-1} \frac{n-1}{n} \frac{n+1}{n}$ which is just

$$\left(\frac{1}{2}\right)\left(\frac{3}{2}\right)\left(\frac{2}{3}\right)\left(\frac{4}{3}\right)\left(\frac{3}{4}\right)\left(\frac{5}{4}\right) \cdots \left(\frac{n-2}{n-1}\right)\left(\frac{n}{n-1}\right)\left(\frac{n-1}{n}\right)\left(\frac{n+1}{n}\right)$$

so we cancel all terms of the kind $(3/2)$ and the very next $2/3$, which yields:

$$\frac{1}{2} \frac{n+1}{n}$$

Example : $\frac{1}{\sqrt{1}+\sqrt{2}} + \frac{1}{\sqrt{2}+\sqrt{3}} + \frac{1}{\sqrt{3}+\sqrt{4}} + \cdots + \frac{1}{\sqrt{n}+\sqrt{n+1}}$

An easy application of rationalizing the denominator yields us a telescope since $1/\sqrt{i} + \sqrt{i+1} = \sqrt{i+1} - \sqrt{i}$. From here we see that the answer is $\sqrt{2} - \sqrt{1} + \sqrt{3} - \sqrt{2} + \sqrt{4} - \sqrt{3} + \cdots + \sqrt{n+1} - \sqrt{n} = \sqrt{n+1} - 1$

Example : (Digits of massive sums) What are the last two digits of the sum

$$1! + 2! + \cdots + 2026!$$

By $10!$, we see that $10!$ and all above factorials have last two digits as 0, so they do not contribute to the last two digits of the sum. We need only calculate $1! + 2! + \cdots + 9!$ to find the digits. $1!$ modulo 100 is 1. $2!$ modulo 100 is 2. $3!$ modulo 100 is 6, $4!$ modulo 100 is 24 but $5!$ modulo 100 is $120 \equiv 20 \pmod{100}$. $6! \equiv 6 \cdot 20 \equiv 20 \pmod{100}$. $7! = 7 \cdot 20 \equiv 40 \pmod{100}$, $8! = 8 \cdot 40 \equiv 20 \pmod{100}$ and $9! = 9 \cdot 20 \equiv 80 \pmod{100}$. Just sum these up to get $1 + 2 + 6 + 24 + 20 + 20 + 40 + 20 + 80 = 213 \equiv 13 \pmod{100}$ so the last two digits are 13.

Note: This is admittedly not a telescoping problem, but a nice application of modular arithmetic on a seemingly large sum.

Example : $\sum_{k=1}^n (k^2 + 1)k!$

Method 1: We write $(k^2 + 1)k! = ((k + 1 - 1)^2 + 1)k! = ((k + 1)^2 + 2 - 2(k + 1))k! = (k + 1)(k + 1)! + 2k! - 2(k + 1)!$. Now write it as a sum:

$$\sum_{k=1}^n (k + 1)(k + 1)! + 2 \sum_{k=1}^n (k! - (k + 1)!)$$

The first term is of the form

$$2(2!) + 3(3!) + \cdots + (n + 1)(n + 1)!$$

which gives us (by way of an earlier telescoping trick) $(n + 2)! - 2$. The second term is of the form

$$2[(1! - 2!) + 2! - 3! + \cdots + n! - (n + 1)!]$$

whence we see the answer is

$$(n + 2)! - 2 + 2 - 2(n + 1)! = (n + 2)! - 2(n + 1)!$$

Method 2: *This ought to be a general scheme, so listen carefully:* We wish to write $(k^2 + 1)k!$ in the form of $f(k + 1) - f(k)$ of some form, to make our telescope idea easier. We need to find that form. Let us look at

$$k^2 + 1 = k(k + 1) - (k - 1) \implies$$

$$(k^2 + 1)k! = k(k + 1)! - (k - 1)k!$$

whence we have found our $f(k) = k(k + 1)!$. Whence we see that $\sum_{k=1}^n (k^2 + 1)k! = \sum_{k=1}^n f(k) - f(k - 1) = f(1) - f(0) + f(2) - f(1) + \cdots + f(n) - f(n - 1) = f(n) - f(0)$, you get the idea. **(Rather important scheme, and is widely applicable to a lot of problems)**

Example : (A very simple, yet very interesting presentation) Evaluate $(1 + x)(1 + x^2)(1 + x^4) \cdots (1 + x^{2^n})$

We note that, if we multiply this by $1 - x$, we get $(1 - x^2)(1 + x^2)(1 + x^4) \cdots = (1 - x^4)(1 + x^4) \cdots = (1 - x^{2^n})(1 + x^{2^n}) = 1 - x^{2^{n+1}}$. So we have $(1 + x)(1 + x^2) \cdots = (1 - x^{2^{n+1}})/(1 - x)$

Example : Sophie Germain Sum: $\sum_{k=1}^n \frac{2k}{k^4 + k^2 + 1}$

Note that $k^4 + k^2 + 1$ can be written as $x^2 + x + 1$ for $k^2 = x$. Now look at $(x - 1)(x^2 + x + 1) = x^3 + x^2 + x - x^2 - x - 1 = x^3 - 1$. So we have $(k^4 + k^2 + 1) = (k^6 - 1)/(k + 1)(k - 1) = (k^3 - 1)(k^3 + 1)/(k + 1)(k - 1)$. Due to the identity $x^3 + 1 = (x + 1)(x^2 - x + 1)$, we have finally:

$$\begin{aligned} (k^6 - 1)/((k + 1)(k - 1)) &= (k^3 - 1)(k^3 + 1)/((k + 1)(k - 1)) = \\ (k - 1)(k^2 + k + 1)(k + 1)(k^2 - k + 1)/(k + 1)(k - 1) &= (k^2 + k + 1)(k^2 - k + 1) \end{aligned}$$

So the sum now reads:

$$\sum_{k=1}^n \frac{2k}{(k^2 + k + 1)(k^2 - k + 1)}$$

whence we can write the numerator $2k = (k^2 + k + 1) - (k^2 - k + 1)$. This immediately gives us

$$\sum_{k=1}^n \frac{(k^2 + k + 1) - (k^2 - k + 1)}{(k^2 + k + 1)(k^2 - k + 1)} = \sum_{k=1}^n \left(\frac{1}{k^2 - k + 1} - \frac{1}{k^2 + k + 1} \right)$$

Now, we still are not done, and it still appears intractable. How do we evaluate *this* sum now? Is this a hidden telescope? Checking for $k = 1$ tells us:

$$1/(1^2 - 1 + 1) - 1/(1^2 + 1 + 1) = 1/1 - 1/3 = 2/3$$

For $k = 2$:

$$1/(4 - 2 + 1) - 1/(4 + 2 + 1) = 1/3 - 1/7$$

For $k = 3$:

$$1/(9 - 3 + 1) - 1/(9 + 3 + 1) = 1/7 - 1/13$$

So we see that it is indeed a telescope, but how do we realise this formulaically? the idea is that, $1/(k^2 + k + 1)$ cancels $1/((k + 1)^2 - (k + 1) + 1)$. Quick evaluation of the latter actually does indeed give us $1/(k^2 + k + 1)$, so we indeed have a telescope. So we have the final formulaic form of the sum as

$$\sum_{k=1}^n \left(\frac{1}{k^2 - k + 1} - \frac{1}{(k + 1)^2 - (k + 1) + 1} \right)$$

This is the definitive telescope, and now we can evaluate it as

$$1/(1^2 - 1 + 1) - 1/((n+1)^2 - (n+1) + 1)$$

which is just

$$1 - 1/(n^2 + n + 1)$$

Example : Factorial sum: $\sum_{k=1}^n (k^3 + 3k^2 + 2k + 1)k!$

Let us look at the polynomial $k^3 + 3k^2 + 2k + 1$. ok this seems to be a dead end problem.